Quantum Information in Photonic systems

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Abstract

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Part I
Photonics Prerequesites

1 Foundations

1.1 Source free Maxwell's Equations as Wave equations

Let us begin with sourceless Maxwell's equations

$$1)\ \nabla.\boldsymbol{E} = 0, \qquad 2)\ \nabla.\boldsymbol{B} = 0, \qquad 3)\ \nabla \times \boldsymbol{E} = -\ \partial_t \boldsymbol{B}, \qquad 4)\ \nabla \times \boldsymbol{B} = \frac{1}{c^2}\ \partial_t \boldsymbol{E}. \tag{1}$$

Combining the curl of 3) with 1) and 4), we can derive, and similarly, combining the curl of 4) with 2) and 3), we can derive

$$\boxed{\nabla^2 \boldsymbol{E} = \frac{1}{c^2}\ \partial_t^2 \boldsymbol{E} \quad \text{with} \quad \nabla.\boldsymbol{E} = 0, \quad \text{and} \quad \nabla^2 \boldsymbol{B} = \frac{1}{c^2}\ \partial_t^2 \boldsymbol{B} \quad \text{with} \quad \nabla.\boldsymbol{B} = 0.} \tag{2}$$

These equation show that both the electric field $\boldsymbol{E}$ and magnetic field $\boldsymbol{B}$ satisfies a wave equation.

Let us now introduce the vector potential $\boldsymbol{A}$. From 2) we get, $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, combining it with 3) we get $\nabla \times (\boldsymbol{E} + \partial_t \boldsymbol{A}) = 0$, and that gives

$$\boldsymbol{E} = -\partial_t \boldsymbol{A} - \nabla\phi. \tag{3}$$ eq:EinA

Under the gauge transformations

$$\boldsymbol{A} \to \boldsymbol{A}\ +\ \nabla\Lambda \quad \text{and} \quad \phi \to \phi\ -\ \partial_t \Lambda \tag{4}$$

both $\boldsymbol{B}$ and $\boldsymbol{E}$ are invariant. Using this gauge freedom, we would like to impose the

$$\text{Coulomb Gauge:} \quad \nabla.\boldsymbol{A} = 0. \tag{5}$$

Putting this in equation (3) and using 1), we get $\nabla.\boldsymbol{E} = -\nabla^2\phi = 0$. This implies that the scalar potential ϕ satisfies the Laplace's equation and it is an harmonic function. A harmonic function that is bounded on all of space or periodic in a box is constant in space i.e. $\phi = c(t)$. Perform a gauge transformation with $\Lambda = \int dt\ c(t)$, this sets $\phi = 0$ without violating the Coulomb gauge condition. Note that this would not be possible in the presence of a source as $\nabla^2\phi = -\ \tilde{\rho}$ where $\tilde{\rho}$ is the charge density. Finally, plugging the expression of $\boldsymbol{E}$ and $\boldsymbol{B}$ in terms of $\boldsymbol{A}$ in 4), we get

$$\boxed{\boldsymbol{E} = -\partial_t \boldsymbol{A}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}, \qquad \nabla^2 \boldsymbol{A} = \frac{1}{c^2}\ \partial_t^2 \boldsymbol{A}, \quad \text{with} \quad \nabla.\boldsymbol{A} = 0 \quad \text{and} \quad \phi = 0} \tag{6}$$

These equations show that in the coulomb gauge, the vector potential also satisfies the same wave equation.

1.2 Classical Solutions to the Wave Equations

In this section we derive the solutions for the Wave equation. We will impose the Coulomb gauge condition $\nabla.\boldsymbol{A} = 0$. Let us pick a separable ansatz for the vector potential, $\boldsymbol{A} = \boldsymbol{f}(\boldsymbol{r})\ T(t)$. As we will see shortly this ansatz and imposing boundedness on $T(t)$ is essentially only looking at monochromatic light. Plugging this into the wave equation, we get

$$T(t)\ \nabla^2 \boldsymbol{f}(\boldsymbol{r}) = \frac{1}{c^2}\ \boldsymbol{f}(\boldsymbol{r})\ T''(t)\ \implies\ \frac{\nabla^2\ f_a(\boldsymbol{r})}{f_a(\boldsymbol{r})} = \frac{T''(t)}{c^2\ T(t)} \tag{7}$$

To get the equation above we have divided the wave equation with the ansatz. The LHS of this equation is only $\boldsymbol{r}$ dependent and RHS is only t dependent. Hence both sides must be equal to a constant λ and we get

$$T''(t) = \lambda\ c^2\ T(t), \qquad \nabla^2 \boldsymbol{f}(\boldsymbol{r}) = \lambda\ \boldsymbol{f}(\boldsymbol{r}). \tag{8}$$

The second equation above is the Helmholtz equation. Let us multiply this equation with $\boldsymbol{f}^*$ and integrate over the space

$$\int d^3r\ \boldsymbol{f}(\boldsymbol{r})^* \left(\nabla^2 \boldsymbol{f}(\boldsymbol{r}) - \lambda\ \boldsymbol{f}(\boldsymbol{r})\right) = 0 \quad \implies \quad \lambda = -\frac{\int d^3r\ |\boldsymbol{f}(\boldsymbol{r})|^2}{\int d^3r\ \left(\partial_a \boldsymbol{f}_b(\boldsymbol{r})\right)\left(\partial_a \boldsymbol{f}_b(\boldsymbol{r})\right)^*} < 0. \tag{9}$$

We have integrated by parts to get the RHS above.

General solution to the wave equation: The temporal part of the solution is

$$T(t) = a\ e^{iwt} + b\ e^{-iwt}, \quad \text{with} \quad \lambda = -\frac{w^2}{c^2} \quad \text{and} \quad w > 0 \tag{10}$$

The solution oscillates with single frequency w i.e. it is monochromatic. Similarly for the spacial part, we write

$$\boldsymbol{f_k}(\boldsymbol{r}) = \boldsymbol{f}(\boldsymbol{k})\ e^{i\boldsymbol{k}.\boldsymbol{r}} \quad \text{with} \quad \lambda = -|\boldsymbol{k}|^2. \tag{11}$$

The combined solutions are of the form

$$\boldsymbol{A_k}(\boldsymbol{r}, t) = \boldsymbol{f}_+(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}+w_{\boldsymbol{k}}t)} + \boldsymbol{f}_-(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}-w_{\boldsymbol{k}}t)} \quad \text{with} \quad w_{\boldsymbol{k}} = c\ |\boldsymbol{k}|. \tag{12}$$

A general solution would be a linear combination of $\boldsymbol{A_k}$s

$$\boldsymbol{A_k}(\boldsymbol{r}, t) = \int \frac{d^3k}{(2\pi)^3} \left[\boldsymbol{f}_+(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}+w_{\boldsymbol{k}}t)} + \boldsymbol{f}_-(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}-w_{\boldsymbol{k}}t)}\right] \quad \text{with} \quad w_{\boldsymbol{k}} = c\ |\boldsymbol{k}|. \tag{13}$$

We require $\boldsymbol{A}$ to be a real field and imposing this condition, we get $\boldsymbol{f}_\pm(-\boldsymbol{k})^* = \boldsymbol{f}_\mp(\boldsymbol{k})$. The Coulomb gauge condition gives $\boldsymbol{k}.\boldsymbol{f}_\pm(\boldsymbol{k}) = 0$. This implies $\boldsymbol{f}_\pm(\boldsymbol{k})$ can only point in the plane transverse to $\boldsymbol{k}$, a vector in this 2d plane, pick two orthonormal complex unit vectors $\boldsymbol{\epsilon}_a(\mathbf{k})$ with $a \in \{+, -\}$. These are the two polarization directions and the general solution is

$$\boldsymbol{A}(\boldsymbol{r}, t) = \int \frac{d^3k}{(2\pi)^3} \sum_{a=\pm} \mathcal{N}_{\boldsymbol{k}}\ \left[\boldsymbol{\epsilon}_a(\boldsymbol{k})\ \alpha_a(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}-w_{\boldsymbol{k}}t)} + c.c.\right], \qquad \mathcal{N}_{\boldsymbol{k}} = \sqrt{\frac{\hbar}{2\varepsilon_0 w_{\boldsymbol{k}}}}. \tag{14}$$

Here $\boldsymbol{k}.\boldsymbol{\epsilon}_a(\boldsymbol{k}) = 0$. The complex numbers $\alpha_a(\boldsymbol{k})$ are arbitrary: they are the "initial data" of the classical field. The constant $\mathcal{N}_{\boldsymbol{k}}$ is pure bookkeeping at this stage. Classically, any constant would do, and $\hbar$ appears only so that later $|\alpha|^2$ will count photons.

Using the solution above, we can now write down the electric field in the Coulomb gauge $\boldsymbol{E} = -\partial_t \boldsymbol{A}$. We write $\boldsymbol{E} = \boldsymbol{E}^{(+)} + \boldsymbol{E}^{(-)}$

$$\boldsymbol{E}^{(+)}(\boldsymbol{r}, t) = i \int \frac{d^3k}{(2\pi)^3} \sum_{a=\pm} \sqrt{\frac{\hbar w_{\boldsymbol{k}}}{2\epsilon_0}}\ \boldsymbol{\epsilon}_a(\boldsymbol{k})\ \alpha_a(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}-w_{\boldsymbol{k}}t)}, \qquad \boldsymbol{E}^{(+)} = \left(\boldsymbol{E}^{(-)}\right)^*. \tag{15}$$

eq:genE+sol

Using the identity $\nabla \times (\boldsymbol{v}\ e^{i\boldsymbol{q}.\boldsymbol{x}}) = i(\boldsymbol{q} \times \boldsymbol{v})\ e^{i\boldsymbol{q}.\boldsymbol{x}}$ and $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, we get

$$\boldsymbol{B}^{(+)}(\boldsymbol{r}, t) = i \int \frac{d^3k}{(2\pi)^3} \sum_{a=\pm} \sqrt{\frac{\hbar}{2\epsilon_0 w_{\boldsymbol{k}}}}\ (\boldsymbol{k} \times \boldsymbol{\epsilon}_a(\boldsymbol{k}))\ \alpha_a(\boldsymbol{k})\ e^{i(\boldsymbol{k}.\boldsymbol{r}-w_{\boldsymbol{k}}t)}, \qquad \boldsymbol{B}^{(+)} = \left(\boldsymbol{B}^{(-)}\right)^*. \tag{16}$$

We can plug this in the expression for the Hamiltonian

$$H = \frac{\varepsilon_0}{2} \int d^3x\ \left(|\boldsymbol{E}|^2 + c^2\ |\boldsymbol{B}|^2\right) = \int \frac{d^3k}{(2\pi)^3}\ \sum_a \hbar w_{\boldsymbol{k}}\ |\alpha_a(\boldsymbol{k})|^2. \tag{17}$$

The normalization constant $\mathcal{N}_{\boldsymbol{k}}$ was chosen to get this clean expression.

1.3 Beams

A beam is a field that (a) carries its energy predominantly in one direction, which we call z, and (b) is confined transversely to a finite width w_0, instead of filling all of space like a plane wave. Hence, we first choose coordinates adapted to a preferred axis, splitting $\boldsymbol{k} = (\boldsymbol{k}_\perp, k_z)$ and $\boldsymbol{r} = (\boldsymbol{r}_\perp, z)$, because that is the natural language for a beam.

Angular Spectrum Representation: We will first rewrite the general solution in a form where it will be easier for us to interpret it as beam.

$$\boldsymbol{E}^{(+)}(\boldsymbol{r},t) = \int \frac{d^2k_\perp}{(2\pi)^2} \int \frac{dk_z}{2\pi}\ \tilde{\boldsymbol{E}}(\boldsymbol{k}_\perp, k_z)\ e^{i(\boldsymbol{k}_\perp.\boldsymbol{r}_\perp\ +\ k_z z\ -\ w_{\boldsymbol{k}} t)} \quad \text{with} \quad w_{\boldsymbol{k}}^2 = c^2\left(\boldsymbol{k}_\perp^2 + k_0^2\right) \tag{18}$$

here $\tilde{\boldsymbol{E}} = \sum_{a=\pm}\ \sqrt{\frac{\hbar w_{\boldsymbol{k}}}{2\epsilon_0}}\ \boldsymbol{\epsilon}_a(\boldsymbol{k})\ \alpha_a(\boldsymbol{k})$. Insert a identity $1 = \int_0^\infty dw\ \delta\left(w - w_k\right)$, into the integral

$$\boldsymbol{E}^{(+)}(\boldsymbol{r},t) = \int_0^\infty \frac{dw}{2\pi}\ e^{-iwt} \int \frac{d^2k_\perp}{(2\pi)^2} \int dk_z\ \delta\left(w - c\sqrt{\boldsymbol{k}_\perp^2 + k_0^2}\right) \tilde{\boldsymbol{E}}\ e^{i(\boldsymbol{k}_\perp.\boldsymbol{r}_\perp\ +\ k_z z)} \tag{19}$$

Under the δ function we replaced w_k by w. Provided $|\boldsymbol{k}_\perp| \leq \omega/c$, the delta function can be rewritten as

$$\delta(w - w_k) = \frac{w}{c^2\ k_z(w)}\Big[\delta(k_z - k_z(w)) + \delta(k_z + k_z(w))\Big], \quad \text{with} \quad k_z(w) = \sqrt{\frac{w^2}{c^2} - |\boldsymbol{k}_\perp|^2} \tag{20}$$

Integrating k_z, we get

$$\boldsymbol{E}^{(+)} = \int_0^\infty \frac{dw}{2\pi} e^{-iwt} \left[\underbrace{\int \frac{d^2k_\perp}{(2\pi)^2} \tilde{\boldsymbol{E}}^f(\boldsymbol{k}_\perp)\ e^{i(\boldsymbol{k}_\perp.\boldsymbol{r}_\perp +\ z k_z(w))}}_{\boldsymbol{E}_f^{(+)}(\boldsymbol{r},w)\text{: Forward branch}} + \underbrace{\int \frac{d^2k_\perp}{(2\pi)^2} \tilde{\boldsymbol{E}}^b(\boldsymbol{k}_\perp)\ e^{i(\boldsymbol{k}_\perp.\boldsymbol{r}_\perp -\ z k_z(w))}}_{\boldsymbol{E}_b^{(+)}(\boldsymbol{r},w)\text{: Backward branch}} \right] \tag{21}$$

with

$$\tilde{\boldsymbol{E}}^f(\boldsymbol{k}_\perp) = \frac{w\ \tilde{\boldsymbol{E}}(\boldsymbol{k}_\perp, k_z(w))}{c^2\ k_z(w)}, \qquad \tilde{\boldsymbol{E}}^b(\boldsymbol{k}_\perp) = \frac{w\tilde{\boldsymbol{E}}(\boldsymbol{k}_\perp, -k_z(w))}{c^2\ k_z(w)}.$$

For monochromatic light i.e. fixed w, the forward and backward branch never mix. In a homogeneous medium each plane wave evolves independently, so forward and backward amplitudes are separately conserved. If the backward amplitude is zero at one plane, it stays zero for all z. Mixing happens only where the medium changes: interfaces, mirrors, gratings, density fluctuations. Those generate reflected, backward waves. Note that this is just a rewriting of the same general solution 15, no approximations have gone in yet.

Let us study the monochromatic, forward propagating solution

$$\boldsymbol{E}_f^{(+)}(\boldsymbol{r},w) = \int \frac{d^2k_\perp}{(2\pi)^2}\ \tilde{\boldsymbol{E}}^f(\boldsymbol{k}_\perp)\ e^{i\left(\boldsymbol{k}_\perp.\boldsymbol{r}_\perp +\ z\ \sqrt{k^2 - |\boldsymbol{k}_\perp|^2}\right)}, \quad w = k\ c. \tag{22}$$

For a beam moving along the z direction, we have $\boldsymbol{k} = (\boldsymbol{k}_\perp, k_0)$ with $w_{\boldsymbol{k}}^2 = c^2\left(\boldsymbol{k}_\perp^2 + k_0^2\right)$. We change

the variables from $k_0 \to w_{\boldsymbol{k}}$ and redefine

$$\frac{dw_{\boldsymbol{k}}}{dk_0} = c^2 \frac{k_0}{w_{\boldsymbol{k}}} = c\cos\theta \approx c\left(1 - \frac{\theta^2}{2}\right), \qquad \alpha_a(\boldsymbol{k}) = \sqrt{c\cos\theta}\ \beta_a(\boldsymbol{k}) \approx \sqrt{c}\ \beta_a(\boldsymbol{k})\left(1 - \frac{\theta^2}{4}\right), \tag{23}$$

$$w_{\boldsymbol{k}} = c\sqrt{|\boldsymbol{k}_\perp|^2 + k_0^2} \approx c\ k_0\left(1 + \frac{\theta^2}{2}\right) \tag{24}$$

Here θ is the angle between $\boldsymbol{k}$ and the z direction the paraxial approximation is $\theta \ll 1$.

$$\boldsymbol{E}^{(+)}(\rho, z, t) = i\int \frac{d^2\boldsymbol{k}_\perp\ dw_{\boldsymbol{k}}}{(2\pi)^3} \sum_{a=\pm} \sqrt{\frac{\hbar\ k_0}{2\epsilon_0}}\ \boldsymbol{\epsilon}_a(\boldsymbol{k}_\perp,\ w_{\boldsymbol{k}})\ \beta_a(\boldsymbol{k}_\perp,\ w_{\boldsymbol{k}})\ e^{i(\boldsymbol{k}_\perp.\rho\ +\ k_0 z\ -\ w_{\boldsymbol{k}}t)} \tag{25}$$

2 Density Matrix Formulation of Skyrmions

2.1 Coherency matrix to the density matrix

The field resolved Stokes parameters are defined from field intensities. For a paraxial, quasi-monochromatic beam has two traverse components represented by two complex fields packaged in a *Jones vector* $E = (E_x(\boldsymbol{r}),\ E_y(\boldsymbol{r}))^T$ with $\boldsymbol{r} = (x, y)$. Let us define *coherency matrix*

$$J(\boldsymbol{r}) = \langle E\ E^\dagger\rangle = \begin{pmatrix} \langle|E_x|^2\rangle & \langle E_x E_y^*\rangle \\ \langle E_x^* E_y\rangle & \langle|E_y|^2\rangle \end{pmatrix} = \frac{1}{2}\left(S_0(\boldsymbol{r})\ \mathbb{I} + S_i(\boldsymbol{r})\ \sigma_i\right). \tag{26}$$

The *Stokes parameter* are defined by

$$S_i(\boldsymbol{r}) = \mathrm{Tr}\left[\sigma_i\ J(\boldsymbol{r})\right], \quad \text{with} \quad \sigma_0 = \mathbb{I}, \tag{27}$$

$$\boldsymbol{S}(\boldsymbol{r}) = \left(\langle|E_x|^2 + |E_y|^2\rangle,\ 2\ \mathrm{Re}\langle E_x^* E_y\rangle,\ 2\ \mathrm{Im}\langle E_x^* E_y\rangle,\ \langle|E_x|^2 - |E_y|^2\rangle\right). \tag{28}$$

We can now define a bona fide density matrix

$$\rho(\boldsymbol{r}) = \frac{J(\boldsymbol{r})}{\mathrm{Tr}J(\boldsymbol{r})} = \frac{1}{\langle|E_x|^2 + |E_y|^2\rangle}\begin{pmatrix} \langle|E_x|^2\rangle & \langle E_x E_y^*\rangle \\ \langle E_x^* E_y\rangle & \langle|E_y|^2\rangle \end{pmatrix} = \frac{1}{2}\left(\mathbb{I} + s_i(\boldsymbol{r})\ \sigma_i\right), \quad s_i(\boldsymbol{r}) = \mathrm{Tr}\left[\sigma_i\ \rho(\boldsymbol{r})\right] \tag{29}$$

The normalized Stokes vector S_i/S_0 is the Bloch vector s_i with $i = 1, 2, 3$. The Bloch vector is

$$\boldsymbol{s}(\boldsymbol{r}) = \left(\frac{2\ \mathrm{Re}\langle E_x^* E_y\rangle}{\langle|E_x|^2 + |E_y|^2\rangle},\ \frac{2\ \mathrm{Im}\langle E_x^* E_y\rangle}{\langle|E_x|^2 + |E_y|^2\rangle},\ \frac{\langle|E_x|^2 - |E_y|^2\rangle}{\langle|E_x|^2 + |E_y|^2\rangle}\right). \tag{30}$$

eq:bloch_vec

We can define the *degree of polarization* (DOP)

$$P(\boldsymbol{r}) = |\boldsymbol{s}(\boldsymbol{r})| = \sqrt{s_1(\boldsymbol{r})^2 + s_2(\boldsymbol{r})^2 + s_3(\boldsymbol{r})^2} = \sqrt{\frac{S_1(\boldsymbol{r})^2 + S_2(\boldsymbol{r})^2 + S_3(\boldsymbol{r})^2}{S_0^2}} = \sqrt{1 - 4\ \det\rho} \tag{31}$$

which is the length of the Bloch vector and *purity*

$$\mathrm{Tr}\rho(\boldsymbol{r})^2 = \frac{1}{2}\left(1 + P(\boldsymbol{r})^2\right) = 1 - 2\ \det\rho, \quad \text{with} \quad \det\rho = \frac{\langle|E_x|^2\rangle\ \langle|E_y|^2\rangle - |\ \langle E_x E_y^*\rangle\ |^2}{\langle|E_x|^2 + |E_y|^2\rangle}. \tag{32}$$

which indicates of the state is pure. The pure state has unit Bloch vector/purity, and $\det(\rho) = 0$ but

$$\frac{\langle|E_x|^2 + |E_y|^2\rangle}{4}\ \geq\ \langle|E_x|^2\rangle\ \langle|E_y|^2\rangle - |\ \langle E_x E_y^*\rangle\ |^2\ \geq\ 0 \tag{33}$$

The averaging $\langle.\rangle$ is the source of mixedness of the density matrix without it $\det\rho = 0$. The saturation of the lower bound happens when the state is pure and the upper bound is saturated when the state is maximally mixed.

2.2 Topology: Fiber bundles defined by $\rho(\boldsymbol{r})$

We can write the spectral decomposition of ρ as follows

$$\rho(\boldsymbol{r}) = \lambda_+(\boldsymbol{r})\,|\psi_+(\boldsymbol{r})\rangle\,\langle\psi_+(\boldsymbol{r})| \;+\; \lambda_-(\boldsymbol{r})\,|\psi_-(\boldsymbol{r})\rangle\,\langle\psi_-(\boldsymbol{r})|\,, \quad \text{with} \quad \lambda_\pm(\boldsymbol{r}) = \frac{1 \pm P(\boldsymbol{r})}{4} \tag{34}$$

$$= \frac{\lambda_+(\boldsymbol{r})}{2}\,(\mathbb{I} + \hat{s}(\boldsymbol{r}).\boldsymbol{\sigma}) + \;\frac{\lambda_-(\boldsymbol{r})}{2}\,(\mathbb{I} - \hat{s}(\boldsymbol{r}).\boldsymbol{\sigma}) \quad \text{with} \quad \hat{s}(\boldsymbol{r}) = \frac{\boldsymbol{s}(\boldsymbol{r})}{P(\boldsymbol{r})} \tag{35}$$

We can define

$$\Pi = |\psi_+\rangle\,\langle\psi_+| = \frac{1}{2}\,(\mathbb{I} + \hat{s}(\boldsymbol{r}).\boldsymbol{\sigma})\,, \qquad \tilde{\Pi} = |\psi_-\rangle\,\langle\psi_-| = \frac{1}{2}\,(\mathbb{I} - \hat{s}(\boldsymbol{r}).\boldsymbol{\sigma})$$

as the projector onto the $\lambda_\pm$ eigenspaces. This density matrix, for $P < 1$ is mixed and has rank-2, for $P = 1$, it is pure, $\lambda_- = 0$, and it has rank-1. The projectors $\Pi(\boldsymbol{r})/\tilde{\Pi}(\boldsymbol{r})$ divide the $\mathbb{C}^2$ into two eigenspaces at each $\boldsymbol{r}$

$$\mathbb{C}^2 = \underbrace{\text{span}_{\mathbb{C}}\{|\psi_+\rangle\,(\boldsymbol{r})\}}_{l(\boldsymbol{r})\;:=\;\lambda_+\;\text{eigenvalue}} \;\oplus\; \underbrace{\text{span}_{\mathbb{C}}\{|\psi_-(\boldsymbol{r})\rangle\}}_{\tilde{l}(\boldsymbol{r})\;:=\;\lambda_-\;\text{eigenvalue}}\,, \qquad l(\boldsymbol{r}) \cup \tilde{l}(\boldsymbol{r}) = \mathbb{C}^2, \qquad l(\boldsymbol{r}) \cap \tilde{l}(\boldsymbol{r}) = \{0\} \tag{36}$$

where

$$l(\boldsymbol{r}) := \left\{|\psi\rangle : \Pi(\boldsymbol{r})\,|\psi\rangle = |\psi\rangle\,,\; |\psi\rangle \in \mathbb{C}^2\right\}, \tag{37}$$

$$\tilde{l}(\boldsymbol{r}) := \left\{|\psi\rangle : \tilde{\Pi}(\boldsymbol{r})\,|\psi\rangle = |\psi\rangle\,,\; |\psi\rangle \in \mathbb{C}^2\right\}. \tag{38}$$

Here we have all the ingredients to define the complex line bundle, and sections on it. The rank-1 projector $\Pi(\boldsymbol{r})$ can always be thought of a the density matrix for some fully polarized beam. Note that $\Pi(\boldsymbol{r}) \equiv \rho(\boldsymbol{r})$ when $P(\boldsymbol{r}) = 1$ i.e. when the state is pure.

This $\Pi(\boldsymbol{r})$ contains information of the normalized Bloch vector field $\text{Tr}(\Pi(\boldsymbol{r})\;\boldsymbol{\sigma}) = \hat{s}(\boldsymbol{r})$ and it defines a *complex line bundle*

$$\mathcal{L} := \left\{(\boldsymbol{r},\; l(\boldsymbol{r})) : \boldsymbol{r} \in \mathbb{R}^2\right\}, \quad \pi : (\boldsymbol{r},\; |\psi\rangle) \to \boldsymbol{r}, \quad |\psi\rangle \in l(\boldsymbol{r}). \tag{39}$$

In addition, if we impose the normalization condition on the states, it defines the $U(1)$ fibers at each $\boldsymbol{r}$

$$p(\boldsymbol{r}) := \left\{|\psi\rangle : \Pi(\boldsymbol{r})\,|\psi\rangle = |\psi\rangle\,,\; \langle\psi|\psi\rangle = 1,\; |\psi\rangle \in \mathbb{C}^2\right\}, \tag{40}$$

$$\tilde{p}(\boldsymbol{r}) := \left\{|\psi\rangle : \tilde{\Pi}(\boldsymbol{r})\,|\psi\rangle = |\psi\rangle\,,\; \langle\psi|\psi\rangle = 1,\; |\psi\rangle \in \mathbb{C}^2\right\}. \tag{41}$$

and hence also defines the *Principle $U(1)$-bundle*

$$\mathcal{P} := \left\{(\boldsymbol{r},\; p(\boldsymbol{r})) : \boldsymbol{r} \in \mathbb{R}^2\right\}, \quad \pi : (\boldsymbol{r},\; |\psi\rangle) \to \boldsymbol{r}, \quad |\psi\rangle \in p(\boldsymbol{r}). \tag{42}$$

The topology hidden in the $\rho(\boldsymbol{r})$ is captured by the fiber bundles described above.

Skyrmions: Skyrmions are usually defined for pure states ($P(\boldsymbol{r}) = 1$). The transitional way is to extract the unit Bloch vector from the measured Stokes parameters. This defines a map from the beam plane to the Poincaré sphere

$$\hat{s} : \mathbb{R}^2 \to S^2 \tag{43}$$

here $\mathbb{R}^2$ is the beam plane. An optical skyrmion is a polarization texture whose Stokes vector $\hat{s}(\boldsymbol{r})$ wraps the whole Poincaré sphere an integer number of times as $\boldsymbol{r}$ ranges over the plane. That integer is the skyrmion number n. A subtlety: the beam cross-section is an infinite plane R2, but "wrapping a sphere" is only an integer-valued idea when the domain is also closed, like a sphere. We fix this by requiring the polarization to approach a single constant value far away, $\hat{s}(\boldsymbol{r}) \to s_0$ as $|r| \to \infty$. Then all the points at

infinity can be glued into one single point, turning the plane into a sphere. Adding a single "point at infinity" to the plane turns it into a sphere:

$$\mathbb{R}^2 \cup \{\infty\} = S^2. \tag{44}$$

Wrapping an infinite sheet of paper up into a ball, with all the far-away edges meeting at the top. The requirement $\hat{s}(\boldsymbol{r}) \to s_0$ is what makes this gluing consistent. Without compactification there is no integer; with it, the skyrmion number is well-defined. The following formula then defines the Skyrmion number

$$n = \frac{1}{4\pi} \int \hat{s}.\left(\partial_x \hat{s} \times \partial_x \hat{s}\right) \, dxdy. \tag{45}$$

As we explained earlier for a pure state $\rho(\boldsymbol{r}) \equiv \Pi(\boldsymbol{r})$. Is is possible to rewrite the skyrmion number using only $\Pi(\boldsymbol{r})$ as it contains the normalized Block vector $\hat{s}$.

$$\Pi(\boldsymbol{r}) = \frac{1}{2}\left(\mathbb{I} + \hat{s}(\boldsymbol{r}).\boldsymbol{\sigma}\right) \quad \implies \partial_\mu \Pi = \frac{1}{2}\partial_\mu \hat{s}.\boldsymbol{\sigma} \tag{46}$$

Using $SU(2)$ algebra, we can derive

$$[\partial_x \Pi, \ \partial_y \Pi] = \frac{1}{4}\left[\partial_x.\boldsymbol{\sigma}, \ \partial_y.\boldsymbol{\sigma}\right] = \frac{i}{2}\left(\partial_x \hat{s} \times \partial_x \hat{s}\right).\boldsymbol{\sigma} \tag{47}$$

$$\mathrm{Tr}\left(\Pi\left[\partial_x \Pi, \ \partial_y \Pi\right]\right) = \frac{i}{2} \, \hat{s}.\left(\partial_x \hat{s} \times \partial_x \hat{s}\right) \tag{48}$$

and hence the skyrmion number purely in terms of Π is

$$\boxed{n = \frac{1}{4\pi} \int \hat{s}.\left(\partial_x \hat{s} \times \partial_x \hat{s}\right) \, dxdy = -\frac{i}{2\pi} \int \mathrm{Tr}\left(\Pi\left[\partial_x \Pi, \ \partial_y \Pi\right]\right) \, dxdy} \tag{49}$$

This definition of the skyrmion number in terms of the projector $\Pi(\boldsymbol{r})$ instead of $\rho(\boldsymbol{r})$ is extendable to the mixed states. One subtlety which we will address later in detail is that the topology can be lost if the density matrix $\rho(\boldsymbol{r})$ becomes degenerate at some point $\boldsymbol{r} = \boldsymbol{r}^*$ i.e $P(\boldsymbol{r}^*) = 0$, the projector $\Pi(r)$ cannot detect this as it does not contain information of the DOP field, $P(\boldsymbol{r})$. Hence while tracking the topological content of the mixed states through an evolution along with $\Pi(\boldsymbol{r}, t)$ one must also keep track of $P(\boldsymbol{r}, t)$. All of this is contained in experimentally measured Stokes parameters.

2.2.1 Geometry Underneath : Transition Functions

As explained earlier, the projector $\Pi(\boldsymbol{r})$ defines the Principle $U(1)$ bundle. It is possible to also define a local section using $\Pi(\boldsymbol{r})$, as follows

1. In a neighborhood $U \in \mathbb{R}^2$, solve for $\Pi(\boldsymbol{r})\,|\tilde{\psi}\rangle = |\tilde{\psi}(\boldsymbol{r})\rangle$
2. Normalize the vector: $|\psi(\boldsymbol{r})\rangle = |\tilde{\psi}(\boldsymbol{r})\rangle / \sqrt{\langle\tilde{\psi}(\boldsymbol{r})|\tilde{\psi}(\boldsymbol{r})\rangle}$
3. Fix the $U(1)$ phase by some local conventions e.g. fix the first component to be real.
4. Call this local section, $|e_U(\boldsymbol{r})\rangle$

Define another local section $|e_V(\boldsymbol{r})\rangle$ in a different neighborhood V. In the intersection $U \cap V$ at a the same point the vector $|e_U\rangle$ and $|e_V\rangle$ belong to the same fiber and hence only differ by a $U(1)$ phase

$$|e_U(\boldsymbol{r})\rangle = g_{UV}(\boldsymbol{r})\,|e_V(\boldsymbol{r})\rangle\,, \qquad g_{UV}(\boldsymbol{r}) = \frac{\langle e_U(\boldsymbol{r})|e_V(\boldsymbol{r})\rangle}{|\,\langle e_U(\boldsymbol{r})|e_V(\boldsymbol{r})\rangle\,|} = e^{i\Theta(\boldsymbol{r})} \in U(1). \tag{50}$$

The winding of the phase counts the skyrmion number

$$n = \frac{1}{2\pi} \oint_\gamma d\Theta \in \mathbb{Z}. \tag{51}$$

The loop γ must lie in the overlap $U \cap V$, and it must enclose the obstruction — a zero of a section, a polarization singularity, a point where the eigenline convention breaks.

Example: The Holomorphic Néel hedgehog. This skyrmion profile comes from a pure density matrix. Let us begin by putting complex coordinates on the beam plane

$$\zeta = x + i\ y = r\ e^{i\phi} \tag{52}$$

This space can be compactified by adding $\{\infty\}$ and using complex place to S^2 map. The skyrmion profile is

$$|\psi(\zeta)\rangle = \frac{1}{\sqrt{w^2 + |\zeta|^2}} \begin{pmatrix} w \\ \zeta \end{pmatrix}, \quad \text{with fixed } w > 0. \tag{53}$$

At the origin this it is well defined but at $r \to \infty$ it is multivalent and depends on the direction of approach

$$r \to 0, \quad |\psi(\zeta)\rangle \to \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad r \to \infty, \quad |\psi(\zeta)\rangle \to \begin{pmatrix} 0 \\ e^{i\phi} \end{pmatrix}. \tag{54}$$

After compactification, $\zeta = \infty$ is a single point, yet this formula assigns it a different phase along every incoming ray. The section is discontinuous there — it cannot be defined at that one point. This is the obstruction, made of nothing but the $U(1)$ phase convention. This obstruction can be removed by a local gauge transformation $|\psi(\zeta)\rangle \to e^{-i\phi}\,|\psi(\zeta)\rangle$ but then it would reappear at the origin. The main feature of the skyrmion profile is that it is not possible to define a global gauge convention in which the profile is well defined everywhere. The gauge invariant content from this field is the Bloch vector

$$\hat{s} = \langle\psi(\zeta)|\boldsymbol{\sigma}|\psi(\zeta)\rangle = \left(\frac{2wx}{w^2 + r^2},\ \frac{2wy}{w^2 + r^2},\ \frac{w^2 - r^2}{w^2 + r^2}\right) \tag{55}$$

at $r \to 0$ it points at the north pole (0, 0, 1) and at $r \to \infty$ at the south pole (1, 0, 0). In order to cover the full Bloch sphere we define the skyrmion on two separate patches and together they cover the full sphere

- North patch $U_N = \mathbb{C}\backslash\{\infty\}$ with $|\psi_N(\zeta)\rangle = \left(w^2 + |\zeta|^2\right)^{-1/2} \begin{pmatrix} w \\ \zeta \end{pmatrix} = \frac{\zeta}{|\zeta|}\ |\psi_S(\zeta)\rangle$
- North patch $U_S = \mathbb{C}\backslash\{0\}$ with $|\psi_S(\zeta)\rangle = \left(w^2 + |\zeta|^2\right)^{-1/2} \begin{pmatrix} we^{-i\phi} \\ |\zeta| \end{pmatrix}$

Now it can be checked that $|\psi_S(\zeta)\rangle$ is well defined at $r \to \infty$ but not at $r \to 0$. Now the transition function on the intersection $\mathbb{C}\backslash\{0, \infty\}$ is

$$g_{SN}(\zeta) = \frac{\langle\psi_S(\zeta)|\psi_N(\zeta)\rangle}{|\,\langle\psi_S(\zeta)|\psi_N(\zeta)\rangle\,|} = e^{i\phi} \in U(1) \tag{56}$$

and with $\Theta(\zeta) = \phi$. The winding of the transition function can be extracted by integrating its phase along a closed loop that encloses the obstruction i.e. poles

$$n = \frac{1}{2\pi} \oint_\gamma d(\arg\, g_{SN}) = \frac{1}{2\pi} \int_0^{2\pi} d\phi = 1 \tag{57}$$

2.2.2 Geometry Underneath : Connection and Curvature

The section over the line bundle assigns a vector $|\psi(\boldsymbol{r})\rangle$ at each base point $\boldsymbol{r}$ which belongs to the $+1$ eigenspace of the projector $\Pi(\boldsymbol{r})$ i.e. $\Pi(\boldsymbol{r})\,|\psi(\boldsymbol{r})\rangle = |\psi(\boldsymbol{r})\rangle$. We would like to understand how does the state $|\psi(\boldsymbol{r})\rangle$ change when we move along the base direction. The naive derivative

$$|\partial_\mu \psi(\boldsymbol{r})\rangle \in \mathbb{C}^2 = l(\boldsymbol{r}) \oplus \tilde{l}(\boldsymbol{r}) \tag{58}$$

typically this vector contains the physical and pure gauge changes. It also leaves the fiber direction and has a components along the 0 eigenspace of the projector $\Pi(\boldsymbol{r})$. The normalization condition

$$\langle\psi(\boldsymbol{r})|\psi(\boldsymbol{r})\rangle = 1 \quad \implies \partial_\mu \langle\psi(\boldsymbol{r})|\psi(\boldsymbol{r})\rangle = \langle\partial_\mu\psi(\boldsymbol{r})|\psi(\boldsymbol{r})\rangle + \langle\psi(\boldsymbol{r})|\partial_\mu\psi(\boldsymbol{r})\rangle = 0, \tag{59}$$

imposes that $\langle\psi(\boldsymbol{r})|\partial_\mu\psi(\boldsymbol{r})\rangle = iA_\mu(\boldsymbol{r})$ is purely imaginary and $A(\boldsymbol{r}) = A_\mu(\boldsymbol{r})dx^\mu$ is *Berry connection*. It also implies that the part of $|\partial_\mu\psi(\boldsymbol{r})\rangle$ along $l(\boldsymbol{r})$ is purely along the $U(1)$ gauge direction. The gauge invariant part of $|\partial_\mu\psi(\boldsymbol{r})\rangle$ which entirely lies in $\tilde{l}(\boldsymbol{r})$, can be obtained by subtracting this pure gauge $U(1)$ part

$$|\mathcal{D}_\mu\psi(\boldsymbol{r})\rangle = (\mathbb{I} - \Pi(\boldsymbol{r}))\,|\partial_\mu\psi(\boldsymbol{r})\rangle = |\partial_\mu\psi(\boldsymbol{r})\rangle - \langle\psi(\boldsymbol{r})|\partial_\mu\psi(\boldsymbol{r})\rangle\ \ |\psi(\boldsymbol{r})\rangle \tag{60}$$

$$= |\partial_\mu\psi(\boldsymbol{r})\rangle - iA_\mu(\boldsymbol{r})\,|\psi(\boldsymbol{r})\rangle \tag{61}$$

$$= (\partial_\mu - iA_\mu(\boldsymbol{r}))\,|\psi(\boldsymbol{r})\rangle\,. \tag{62}$$

Note that, under the gauge transformation

$$|\psi(\boldsymbol{r})\rangle \to e^{i\chi}\,|\psi(\boldsymbol{r})\rangle \implies A_\mu(\boldsymbol{r}) \to A_\mu(\boldsymbol{r}) + \partial_\mu\chi \tag{63}$$

Berry connection transform as a $U(1)$ connection. From the connection 1-form $A(\boldsymbol{r}) = A_\mu(\boldsymbol{r})dx^\mu$ we can define the curvature 2-form $F = dA$. The Chern–Weil integrality theorem restricts the global behavior of F. It implies that for any closed oriented surface Σ inside base space, the total flux of the curvature through that surface must be a quantized integer:

$$c_1 = \frac{i}{2\pi}\int_\Sigma F \in \mathbb{Z} \tag{64}$$

This integer in know as the 1st Chern number of the bundle and it is the skyrmion number.

Example: The Holomorphic Néel hedgehog. The connection one form is $A = -i\,\langle\psi|d\psi\rangle$, let us derive it from the Skyrmion profile. First we make a useful change of coordinates $r/(w^2+r^2)^{1/2} = \sin(\Theta/2)$, this choice puts the skyrmion profile into the conventional Block form and $\Theta \in (0,\ \pi)$.

$$|\psi_N(\Theta,\phi)\rangle = \begin{pmatrix}\cos(\Theta/2)\\ \sin(\Theta/2)\ e^{i\phi}\end{pmatrix} \tag{65}$$

this gives us

$$|\partial_\Theta\psi_N\rangle = \frac{1}{2}\begin{pmatrix}-\sin(\Theta/2)\\ \cos(\Theta/2)\ e^{i\phi}\end{pmatrix}, \qquad |\partial_\phi\psi_N\rangle = i\begin{pmatrix}0\\ \sin(\Theta/2)\ e^{i\phi}\end{pmatrix} \tag{66}$$

with this, we have

$$A_N = -i\,(\langle\psi|\partial_\Theta\psi\rangle\,d\Theta + \langle\psi|\partial_\phi\psi\rangle\,d\phi) = \sin^2\frac{\Theta}{2}\,d\phi \tag{67}$$

The connection A_N is also not well defined at $\Theta = \pi$ ($r \to \infty$) as ϕ is degenerate. As $|\psi_S\rangle = e^{i\phi}\,|\psi_N\rangle$ using the transformation rule on the connection

$$A_S = A_N + \partial_\mu\phi\ dx^\mu = A_N + d\phi = \left(1 + \sin^2\frac{\Theta}{2}\right)\,d\phi \tag{68}$$

The connection A_S is also not well defined at $\Theta = 0$ ($r \to 0$) as ϕ is degenerate. The curvature two form is $F = \partial_\mu A_\nu\ dx^\mu \wedge dx^\nu$, then

$$F = \frac{1}{2}\ \sin\Theta\ d\Theta \wedge d\phi. \tag{69}$$

Being a gauge invariant quantity it has the same expression when derived from both the connection 1-forms. Then the Chern number is

$$c_1 = \frac{1}{2\pi}\int_{S^2} F = \frac{1}{4\pi}\int_0^{2\pi} d\phi \int_0^{\pi} \sin\Theta \; d\Theta = \frac{1}{4\pi} \times 2\pi \times 2 = 1 = n. \tag{70}$$

Any loop γ on the base divide the compactified sphere into two caps. Choose a γ that encloses a poles so that on each cap at least on of the forms is well defined. The integral for the Chern number can be divided into the two caps divided by γ

$$c_1 = \frac{1}{2\pi}\left(\int_S F + \int_N F\right) = \frac{1}{2\pi}\oint_\gamma (A_N - A_S) = \frac{1}{2\pi}\int_0^{2\pi} d\phi = 1. \tag{71}$$

As we can see that the last integrand on the RHS, the difference of the two forms precisely boils down to the differential of the phase of the transition function.

3 Evolution of Density Matrix Under a CPTP Channel

Part II
Discord

4 Discord as Gap Between Classical and Quantum Joint/Conditional Entropy

4.1 Two Formulas of Classical Joint and Conditional Entropy

We begin with expression for conditional entropy in terms of joint probability distribution of two random variables $p(a,b)$

$$H(A|B) = \sum_b p(b)\ H(A|B=b) \tag{72}$$

Here $p(b) = \sum_a p(a,b)$, $H(A|B)$ is Shanon's entropy when B is known i.e. conditional entropy and $H(A|B=b)$ is Shanon's entropy when $B=b$. We can rewrite this formula in terms of a joint entropy

$$\begin{aligned}
H(A|B) &= \sum_b p(b)\ H(A|B=b) \\
&= -\sum_b p(b)\ \sum_a p(a|b) \ln p(a|b) \\
&= -\sum_{ab} p(a,b) \left[\ln p(a,b) - \ln p(b)\right] \\
&= -\sum_{ab} p(a,b)\ \ln p(a,b) + \sum_{ab} p(a,b)\ \ln p(b) \\
&= -\sum_{ab} p(a,b)\ \ln p(a,b) + \sum_{b} p(b)\ \ln p(b) \\
&= H(A,B) - H(B).
\end{aligned} \tag{73}$$

We have two classically equivalent formulas for the conditional entropy $H(A|B)$ and hence, also for the joint entropy $H(A|B)$.

$$H(A,B)_s = -\sum_{ab} p(a,b)\ \ln p(a,b) \qquad \to \text{Symmetric form} \tag{74}$$

$$H(A,B)_c = \sum_b p(b)\ H(A|B=b) + H(B) \qquad \to \text{Conditional form} \tag{75}$$

Classically these two formulas are identical, the subscripts will be useful in the next section. In the conditional formula, knowing B does not change the joint distribution $p(a,b)$.

4.2 Quantum Mechanical Generalization

4.2.1 Quantum Measurement

Quantum mechanically, "knowing" B is locally measuring the system B in some orthonormal (pointer) basis and it changes the joint global state of the combines system. A non-selective measurement-"the measurement is done but we don't look at the result"-is described by a dephasing channel. Suppose the joint state of the system is ρ_{AB} and B is being measured in some orthonormal basis $\{|B_k\rangle\}$. In this basis, we can express the state as follows

$$\rho_{AB} = \sum_{kk'} |B_k\rangle \langle B_{k'}| \ \otimes \underbrace{\langle B_k|\rho_{AB}|B_{k'}\rangle}_{A_{kk'}} = \sum_{kk'} |B_k\rangle \langle B_{k'}| \ \otimes A_{kk'} \tag{76}$$

$$= \sum_k p_k\ \rho_A^{(k)} \ \otimes \ |B_k\rangle \langle B_k| + \sum_{k\neq k'} |B_k\rangle \langle B_{k'}| \ \otimes A_{kk'}. \tag{77}$$

Here, $A_{kk} = p_k\ \rho_A^{(k)}$ with $\mathrm{Tr}\rho_A^{(k)} = 1$, and the off-diagonal $A_{kk'}$ are operator valued coherences that mix different $|B_k\rangle$ sectors. It is these coherences that the dephasing channel, defined using $\{|B_k\rangle\}$, sets to

0 i.e. measurement of B disturbs the joint global state of the combined system. Mathematically, the dephasing/non-selective measurement channel $\mathcal{D}_{\{\Pi_k^B\}}(.)$ acts as follows:

$$\rho_{AB} \to \mathcal{D}_{\{\Pi_k^B\}}(\rho_{AB}) := \rho'_{AB} = \sum_k \left(\mathbb{I}^A \otimes \Pi_k^B\right) \ \rho_{AB} \ \left(\mathbb{I}^A \otimes \Pi_k^B\right) \tag{78}$$

$$= \sum_k p_k \ \rho_A^{(k)} \ \otimes \ |B_k\rangle \langle B_k| \tag{79}$$

here $\Pi_k^B = |B_k\rangle \langle B_k|$.

Classical nature of correlations in the post-measurement state: Defining features of classical correlations between two variables when one of them is observed are

1. Values that the conditioning variable takes (i.e $|B_k\rangle$) are perfectly distinguishable.
2. Learning it, updates "your knowledge" of the other system without changing the global state.

The $A-B$ correlations post measurement/dephasing in $\{\Pi_k^B\}$ are of this kind. The $|B_k\rangle$ are orthonormal, perfectly distinguishable states and if we were to selectively measure ρ'_{AB} in these basis, it does not change the global state but just updates your knowledge of A i.e. A is in $\rho_A^{(k)}$.

Note that this is a basis $\{|B_k\rangle\}$ dependent statement as the state ρ'_{AB} takes the form of the classical catalouge-"learn the label k for B, update your knowledge of A, disturb nothing"-only in this basis. We will refer to this "classical catalouge" form of the state as classical-quantum (Classical in B, quantum in A) form. The basis dependence is also clear from the fact that the pre-measurement state ρ_{AB} does not have the classical-quantum form in generic basis. If there exists a measurement basis $\{|\tilde{B}_k\rangle\}$, such that in them

$$\langle \tilde{B}_k|\rho_{AB}|\tilde{B}_{k'}\rangle = 0 \iff \rho_{AB} = \mathcal{D}_{\{\tilde{\Pi}_k^B\}}(\rho_{AB}) \,. \tag{80}$$

Then pre and post measurement states are the same and measurement in this basis does not disturb the joint global state. In this basis, the state has the classical-quantum form:

$$\rho_{AB} = \sum_k \tilde{p}_k \ \tilde{\rho}_A^{(k)} \ \otimes \ |\tilde{B}_k\rangle \langle \tilde{B}_k| \tag{81}$$

with $\langle \tilde{B}_k|\rho_{AB}|\tilde{B}_k\rangle = \tilde{p}_k \ \tilde{\rho}_A^{(k)}$. In general, such a basis does not exit for all joint states and hence measurement of B, generally disturbs the joint global state.

4.2.2 Quantum Generalization

Let us now write the quantum generalization of the two classical formulas for joint entropy. The generalization of the symmetric formula is straight forward, the formula for Shanon entropy $H(A,B)$ will be replaced by the formula for von Neumann entropy $S(\rho_{AB})$ of the joint state:

$$S(A,B)_s = S(\rho_{AB}) \,. \tag{82}$$

In the generalization of the conditional formula, we will replace the post-measurement entropy $H(A|B=b)$ by von Neumann entropy of the post-measurement state of A, $S\left(\rho_{(k)}^A\right)$:

$$S(A,B)_c = S(A|B)_{\{\Pi_k^B\}} + S(B) = \sum p_k \ S\left(\rho_{(k)}^A\right) + S\left(p_k \ |B_k\rangle \langle B_k|\right) = S\left(\rho'_{AB}\right) . \tag{83}$$

The quantum generalization of two formulas for joint entropy which are classically identical compute two different quantities! It can be shown that the von Neumann entropy $S(.)$ is always non-decreasing under dephasing is channel. The positive difference is

$$\delta_{\{\Pi_k^B\}} = S\left(\mathcal{D}_{\{\Pi_k^B\}}(\rho_{AB})\right) - S\left(\rho_{AB}\right) \geq 0 \tag{84}$$

It is clear, that if there exist a basis $\left\{\tilde{\Pi}_k^B\right\}$ in which the state has the classical-quantum form then, the measurement in this basis does not disturb the state and hence $\delta_{\{\tilde{\Pi}_k^B\}} = 0$. If such a basis does not exit, we should minimize over all possible choice of basis, *this minimum positive joint entropy generated by minimizing the measurement-disturbance to the joint state is known as quantum discord*:

$$\boxed{\delta(A:B) = \min_{\{\Pi_k^B\}} \left[S\left(\mathcal{D}_{\{\Pi_k^B\}}(\rho_{AB})\right)\right] - S\left(\rho_{AB}\right).} \tag{85}$$

It will become clear in the next section that discord it also a measure of non-classical $A-B$ correlations.

5 Discord as Gap between Classical and Quantum Mutual Information

5.1 Classical Mutual Information and Its Quantum Generalization

The two formulas for classical mutual information are

$$I(A:B) = H(A) + H(B) - H(A,B) \qquad \rightarrow \text{Symmetric form} \tag{86}$$
$$J(A:B) = H(A) - H(A|B) \qquad \rightarrow \text{Conditional form} \tag{87}$$

The mutual information compute the amount of correlation between the A and B in bits. Classically these are quantities identical. Let us not generalize these to quantum formulas. The generalization of the symmetric formula is again straight forward.

$$I(A:B) = S(A) + A(B) - S(A,B) \tag{88}$$

Note that this formula account both classical + quantum/non-classical correlations. It is also property of the pre-measurement joint state ρ_{AB}. The generalization of the conditional formula

$$J(A:B)_{\{\Pi_k^B\}} = S(A) - S\left(A|\{\Pi_k^B\}\right) = S(\rho_A) - \sum_k p_k \; S\left(\rho_{(k)}^A\right) \tag{89}$$

has the same subtlety as in the conditional/joint entropy, it is basis dependent. The difference is

$$\delta_{\{\Pi_k^B\}} = I(A:B) - J(A:B)_{\{\Pi_k^B\}} \geq 0 \tag{90}$$

is the same quantity defined above and its minimization gives us the discord:

$$\boxed{\delta(A:B) = \min_{\{\Pi_k^B\}} \left[I(A:B) - J(A:B)_{\{\Pi_k^B\}}\right] = I(A:B) - \max_{\{\Pi_k^B\}} \left[J(A:B)_{\{\Pi_k^B\}}\right]} \tag{91}$$

This formula gives us an intuition for discord: *it the minimum amount of $A-B$ correlation destroyed by the measurement of B in any basis.*

5.2 Discord is Non-classical Correlation Beyond Entanglement

A general entangled state is always discordant (has positive discord) but the converse it not true

$$\{\text{Entangled states}\} \subsetneq \{\text{Discordant states}\} \tag{92}$$

5.2.1 Product State: Unentangled and Discordless

Let us begin by observing a **product state**

$$\rho_{AB} = \rho_A \otimes \rho_B.$$

This state LOCC preparable and hence there is no $A-B$ entanglement by definition. Choose any orthonormal set of basis $\left\{|\hat{B}_k\rangle\right\}$ for B, then in this basis the state has the classical-quantum form

$$\rho_{AB} = \sum_k \hat{p}_k \; \rho_A \otimes |\hat{B}_k\rangle\langle\hat{B}_k| \quad \text{with} \quad \rho_{(k)}^A = \rho_A \; \forall \; k. \tag{93}$$

Hence, for a product state.

5.2.2 Separable State: Unentangled but Discordant

Let us now go one step ahead and consider a **separable state** which is defined as a probabilistic mixture of product state,

$$\rho_{AB} = \sum p_k \ \rho_A^k \ \otimes \ \rho_B^k$$

with $p_k \geq 0$ and $\sum p_k = 1$. This also is LOCC preparable and has no entanglement by definition. It is generally not possible to find an orthonormal basis such that $\langle B_l|\rho_B^k|B_{l'}\rangle = 0$ for all k simultaneously with $l \neq l'$.

6 Discord of Gap Between Efficiency of a Quantum and classical Maxwells Demon

6.1 Certainty is fuel

The Maxwell's demon is an entity observing molecules of a gan in a box with a partition in the middle. It sorts the molecules into fast and slow, effectively making one side hot and the other cold. This lowers the entropy of an isolated system violating the second law. It is not possible to extract work by attaching a piston to the partition and letting the gas expand. The Demons job is to acquire the information, sort the gas i.e. lower the entropy which in turn lets you extract work from the system. From this we can infer a physical phenomenon, *lowering entropy or increasing certainty is a "fuel" that can be converted into and extracted as work.* We now make this concrete below.

Extraction of Work in a Thermodynamic Process: Let us formalize this setup. Consider a system in contact with a bath at temperature $T_b = \beta^{-1}$. We would like to estimate how much work can be extracted from this system (coupled to a bath) when it is driven through a process.

We begin with recalling some definitions. The first law of thermodynamics states

$$W_{ext} = Q - \Delta U_s = Q - \Delta \langle H \rangle . \tag{94}$$

The non equilibrium free energy of system is given by

$$F(\rho_s, H) = \langle H \rangle - T \ S(\rho_s) = \mathrm{Tr}(\rho_s H) - T \ S(\rho_s). \tag{95}$$

The mutual information $I(s:b) = S\left(\rho_{sb} \parallel \rho_s \otimes \rho_b\right)$ where $S(\rho\|\sigma) = \mathrm{Tr}\left(\rho \ln \rho - \rho \ln \sigma\right) \geq 0$ is relative entropy which is a measure for difference in the space of density matrices, and $\rho_{s/b}$ are reduced density matrices of the system/bath. We let these systems interact via a unitary coupling for some finite time

$$\rho'_{sb} = U \ \rho_{sb}(0) \ U^\dagger = U \ \left(\rho_s(0) \otimes \gamma_\beta^b\right) \ U^\dagger \tag{96}$$

Here $\rho_{sb}(0)/\rho'_{sb}$ is the joint state before/after coupling is switched on/off. Before coupling, it is an unentangled product state of bath in thermal equilibrium $\gamma_\beta^b = e^{-\beta H_b}/Z$, and $\rho_s(0)$, the state of the system. We can derive how much heat is exchanged during this interaction.

$$Q = T_b \ \Delta S_s - T_B \ \left[S(\rho'_{sb} \parallel \rho'_s \otimes \rho'_b) + S(\rho'_b \parallel \gamma_\beta^b)\right] = T_b \ \Delta S_s - T_b \ \Sigma \tag{97}$$

here ρ'_s/ρ'_b is the reduced density matrix of the system/bath after coupling is switched off. As the relative entropy is always non-negative we conclude that $\Sigma \geq 0$. Combining this with the first law, we get

$$\boxed{W_{ext} = -\Delta\left[U_s - T_b \ S_s\right] + T_b \ \Sigma = -\Delta F_s + T_b \ \Sigma \implies W_{ext} \leq -\Delta F_s.} \tag{98}$$

The change in the free energy of a system in a process is the upper bound on the amount work that can be extracted during this process. The inequality is satisfied when $\Sigma = 0$. Recall that Σ is the total entropy produced during the interaction

$$\Sigma = \beta \ Q + \Delta S_s = \Delta S_b + \Delta S_s = \Delta S_{total}. \tag{99}$$

Demanding $\Delta S_{total} = 0$ implies the process of *interaction must be a reversible process.* In an irreversible process, this is the amount of total entropy produced, and multiplied by T_b, its also the amount of

entropy that could not be converted into work. One particular irreversibility we will be interested in is the decoherence of the system due to measurement in the wrong basis. We can re-express the formula as follows

$$W_{ext} = -\Delta F_s = F(\rho_s(0)) - F\left(\rho_s'\right) = T_b\ S\left(\rho_s(0) \parallel \rho_s'\right). \tag{100}$$

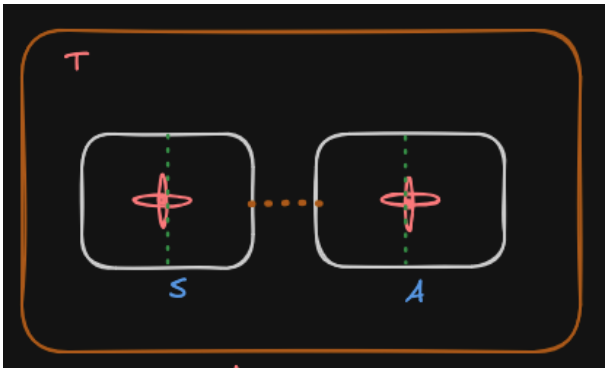
In the process of work extraction the state with certain entropy evolved into a state with more entropy and hence the maximum work that can be extracted when the final state has maximum possible entropy i.e. its in equilibrium at the bath temperature $\rho_s' = \gamma_\beta^s = e^{-\beta H_s}/Z$,[2] hence

$$\boxed{W_{max} = -\Delta F_s = F(\rho_s(0)) - F\left(\gamma_\beta^s\right) = T_b\ S\left(\rho_s(0) \parallel \gamma_\beta^s\right).} \tag{101}$$

The maximum amount of work that can be extracted from a system in state $\rho_s(0)$ is its "distance" from the equilibrium state. This is an process independent statement upper bound.

6.2 Demon as an engine

Let us strip down all the complications of a conventional Maxwell demons problem. Consider the system $\mathcal{S}$ to be a single molecule in a box and it has two states 1) its in the left half of the box, $|L_s\rangle$ 2) its in the right half of the box $|R_s\rangle$. These two states have the same energy hence the Hamiltonian is proportional to identity matrix. There is another physical object which we consider as the register/apparatus $\mathcal{A}$. This is the object where the demon write his knowledge in. We can consider this also to be a molecule with two $|L_a\rangle\,/\,|R_a\rangle$ states. Both of these will be assume to be in contact with a bath at temperature T_b. We



must first distinguish between a classical and quantum demon

- *Classical Demon*: It can measure $\mathcal{A}$ in a fixed $|L_a\rangle\,/\,|R_a\rangle$ basis and act on $\mathcal{S}$ depending on this outcome i.e. LOCC operations.
- *Quantum Demon*: Has global unitary control on combined $\mathcal{S}-\mathcal{A}$ (its a universal quantum computer) and it can perform joint measurements.

The job of the demon is to know the state of $\mathcal{S}$, this certainty about the state is fuel which it will then convert into work according to the formula derived in the last section. A classical demon gains this certainty about the state of $\mathcal{S}$ by measuring $\mathcal{A}$ in some basis, unavoidably dephasing the joint state transforming it into the quantum-classical form. The quantum demon can just rotate the joint state into a product state and measure it without dephasing. We will see that the difference in the work that can be extracted from these two states (disturbed vs undisturbed) is exactly discord.

An engine has a cycle where at the end of the cycle it return to its initial state. Here is the cycle

Step 1: Initial State Consider $\mathcal{S}$ and $\mathcal{A}$ to be in following stats

$$\mathcal{S}: \quad |+\rangle = \frac{1}{\sqrt{2}}\left(|L_s\rangle + |R_s\rangle\right) \quad \text{and} \quad \mathcal{A}: \quad |L_a\rangle\,. \tag{102}$$

[2] here H_s is the post interaction Hamiltonian, the equilibrium is defined with respect to this Hamiltonian

We are unaware of where the $\mathcal{S}$ molecule is and the $\mathcal{A}$ molecule is in a know state. This implies

$$S(\mathcal{S}) = S(\mathcal{A}) = S(\mathcal{S},\mathcal{A}) = I(\mathcal{S}:\mathcal{A}) = 0 \tag{103}$$

Step 2: Create Correaltion The $\mathcal{S}$ and $\mathcal{A}$ is correlated this is considered as premeasurement and is part of the apparatus setup not performed by the demon. We can model this by a CNOT gate with $\mathcal{S}$ as the control then

$$\mathcal{S}-\mathcal{A} : |\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|L_s\ L_a\rangle + |R_s\ R_a\rangle\right) \tag{104}$$

this implies

$$S(\mathcal{S}) = S(\mathcal{A}) = \ln 2, \quad S(\mathcal{S},\mathcal{A}) = 0, \quad I(\mathcal{S}:\mathcal{A}) = 2\ln 2. \tag{105}$$

Step 3: Measure and Extract Work Demons enter here! Let us discuss the action of classical vs quantum demons.

- **Classical Demon:** Measures/dephases $\mathcal{A}$ in some orthonormal basis. A special property of the joint state $\rho_{\mathcal{SA}} = |\Phi^+\rangle\langle\Phi^+|$ and the post ($\mathcal{A}$) measurement state $\rho'_{\mathcal{SA}}$ is that it has the same numerical elements in any orthonormal basis .

$$\rho_{\mathcal{SA}} = \frac{1}{2}\begin{pmatrix}1&0&0&1\\0&0&0&0\\0&0&0&0\\1&0&0&1\end{pmatrix} \quad -\text{dephasing}\rightarrow \quad \rho'_{\mathcal{SA}} = \frac{1}{2}\begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&1\end{pmatrix} \tag{106}$$

  this implies

$$S(\mathcal{S}) = S(\mathcal{A}) = \ln 2, \quad S_s(\mathcal{S},\mathcal{A}) = 2\ln 2, \quad S_c(\mathcal{S},\mathcal{A}) = \ln 2, \quad \delta(\mathcal{S}|\mathcal{A}) = \ln 2. \tag{107}$$

  We an see that 1 bit of correlation is destroyed and the joint entropy has increased by 1 bit. Note that $S_s(\mathcal{S},\mathcal{A})$ is joint entropy of pre measurement state and $S_c(\mathcal{S},\mathcal{A})$ is the joint entropy of the post-measurement state. The difference as explained before is discord. The maximum amount of work that can now be extracted by a classical demon is given by

$$\boxed{W^C_{max} = T_b\ S\left(\rho'_{\mathcal{SA}} \parallel \mathbb{I}/4\right) = T_b\ \left(\ln(4) - S\left(\rho'_{\mathcal{SA}}\right)\right) = T_b \ln 2\,(2-1) = T_b \ln 2} \tag{108}$$

- **Quantum Demon:** Rotates the entangled joint state back into a product state by an unitary U and measure it

$$|\Phi^+\rangle \quad -U_{\mathcal{SA}}\rightarrow \quad |L_sL_a\rangle \tag{109}$$

  this can be done by choosing U to be a combination of CNOT and Hadamard gate. Now measuring both does not disturb/dephase the joint state hence we have

$$S(\mathcal{S}) = S(\mathcal{A}) = 0, \quad S(\mathcal{S},\mathcal{A}) = 0, \quad I(\mathcal{S}:\mathcal{A}) = 0. \tag{110}$$

  Hence, the maximum amount of work that can now be extracted by a quantum demon is given by

$$\boxed{W^Q_{max} = T_b\ S\left(|L_sL_a\rangle \parallel \mathbb{I}/4\right) = T_b\ \left(\ln(4) - S\left(|L_sL_a\rangle\right)\right) = T_b \ln 2\,(2-0) = 2\ T_b \ln 2} \tag{111}$$

The extra work the quantum demon extracts

$$\boxed{W^Q_{max} - W^C_{max} = T_b \ln 2 = T_b\ \delta(\mathcal{S}|\mathcal{A})} \tag{112}$$

is exactly the discord, the amount of correlation destroyed by the classical demon in the process of measuring.

Step 4: Reset Note that, as we explained earlier, the work extraction requires increase in entropy, this also implies that in order to decrease entropy of a state we have put in certain amount of work.

Part III
Appendix